

Predicting Player Spending Behavior and Profitability in Mobile Games

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1. Introduction

The mobile gaming industry has become one of the most profitable segments of the global entertainment market, generating billions of dollars in annual revenue. In recent years, some popular titles have even given rise to a new subindustry, esports, where competitive gaming has evolved into a global cultural and economic phenomenon. Among these games, *Honor of Kings* stands out as one of the most successful and enduring examples, maintaining a massive global player base and consistently generating exceptional revenue. According to third-party reports, *Honor of Kings* achieved an estimated monthly revenue of over 1.2 billion CNY (approximately 168.55 million USD) in the first half of 2025 (Sina Finance, 2025; iTiger News, 2025).

This revenue, however, is highly concentrated: a small percentage of “whale” users contribute a disproportionately large share of total spending. Understanding what drives players to spend in mobile games and the predictability of their spending behavior has become a critical question for game studios seeking to optimize monetization strategies, improve player retention, and forecast revenue. This motivated my project, as I was interested in the economic dynamics behind such large-scale success.

To explore this question, I used a dataset of over 3,000 simulated players from Kaggle that includes detailed behavioral metrics, demographic data, and monetization information. From this dataset, I also created two outputs: a log-transformed continuous spending variable for regression analysis and a binary “whale” indicator identifying the top 5% of spenders for classification. These inputs enable the project to analyze both overall spending behavior and the identification of high-value users.

In this project, I develop machine learning models, including Lasso regression, decision trees, random forests, XGBoost, and logistic regression, to predict player spending and classify whale users. I also use SHAP values to interpret model behavior and identify which features most influence spending outcomes. The results indicate that predicting at the individual level is challenging due to noisy and fragmented behavioral signals, but consistent group-level patterns emerge that can inform monetization strategies. Overall, this project aims to understand the behavioral drivers of spending and assess the feasibility of predictive modeling in a modern mobile gaming environment.

2. Related Work

Recent research on mobile-game monetization highlights several consistent factors that influence both the likelihood and amount of player spending. Early foundational work shows that

engagement structures and monetization mechanics affect purchase behavior (Hamari & Lehdonvirta, 2010; Kim & Lee, 2019). Building on this insight, behavioral studies have identified that specific activity patterns serve as strong predictors of spending. For example, Sifa *et al.* (2015) find that early purchase timing, playtime patterns, and login frequency strongly predict purchase decisions. Similarly, Chen *et al.* (2018) and Zhang *et al.* (2023) further demonstrate that past transaction behavior, especially spending concentration and purchase recency, plays an important role in distinguishing high-value players.

Other research includes demographic and psychological factors. Costes and Bonnaire (2022) find that age, gender, impulsivity, and gaming motives relate to spending habits, though their predictive power is usually weaker than behavioral features. Lee *et al.* (2025) add that psychological dependence may strengthen purchase intention, complementing the behavioral view. From a modeling perspective, Zhang *et al.* (2021) demonstrate that data-driven techniques such as collaborative filtering can forecast spending even in the face of uncertain early user behavior, highlighting the potential of machine learning approaches.

Despite these valuable contributions, several limitations remain in the existing literature. Many studies rely on small-scale surveys or psychological experiments, focus primarily on binary purchase decisions rather than spending amounts, or analyze a single game environment, limiting generalizability. This project aims to address these gaps by developing a generalizable machine learning framework that predicts individual in-app purchase amounts and identifies high-value “whale” users early in the player lifecycle.

Guided by insights from earlier studies, it incorporates engineered features such as *AvgSpendPerSession*, *EngagementScore*, and temporal measures like *DaysSinceLastPurchase*. Although inspired by spending dynamics observed in *Honor of Kings*, the framework is intentionally designed to generalize across genres, countries, and player groups.

3. Data Description

This project uses a structured dataset of 3024 mobile-game players, sourced from Kaggle. After preprocessing and removing observations with missing or invalid values in key behavior or spending fields, the final modeling dataset contains 2734 valid samples. This ensures that all models are trained on complete, consistent observations.

3.1 Dataset Structure

The processed DataFrame includes 62 columns, spanning:

- Demographic attributes: Age, Gender, Country
- Device attribute: Device type
- Gameplay and engagement features: SessionCount, AverageSessionLength, GameGenre

- Behavioral timing features: FirstPurchaseDaysAfterInstall, DaysSinceLastPurchase
- Financial and payment information: PaymentMethod categories, SpendingSegment
- Engineered features designed based on prior literature, including:
 $\text{EngagementScore} = \text{SessionCount} \times \text{AverageSessionLength}$
 → captures total play activity
 $\text{AvgSpendPerSession} = \text{TotalSpend} / \text{SessionCount}$
 → measures spending intensity per visit

After one-hot encoding and numerical filtering, the final modeling matrix used for training contains 56 numeric predictor variables.

3.2 Preprocessing Steps

To prepare the dataset for modeling, the following steps were applied:

- Missing values handled via simple numerical imputation
- Categorical features encoded using one-hot encoding
- Log-transform applied to spending due to heavy right skew
- Generated engineered variables informed by prior literature
- Retained only numeric variables for model training
- Training/testing split: 80/20
- Player segmentation performed through clustering approaches

3.3 Train/Test Split

Using an 80/20 split, the final 2734-row dataset was divided into:

- Training set: 80%
- Test set: 20%

3.4 Target Variables

The project uses two outputs:

1. Regression target: *log-transformed in-app purchase amount*
2. Classification target: *whale indicator*

These two targets correspond to the project's two main objectives:

1. Predict total in-app spending using supervised regression models
2. Identify high-value "whale" users early to support monetization and retention strategies

3.5 Data Visualization and Exploratory Data Findings

To understand the underlying data distributions and behavioral patterns, this project includes a set of 11 exploratory plots, covering:

- Raw vs. log-transformed spending distribution

- Spending differences across genres, countries, and payment methods
- Behavioral predictors and their correlations with spending
- Engineered feature behavior
- Cluster-based player segmentation

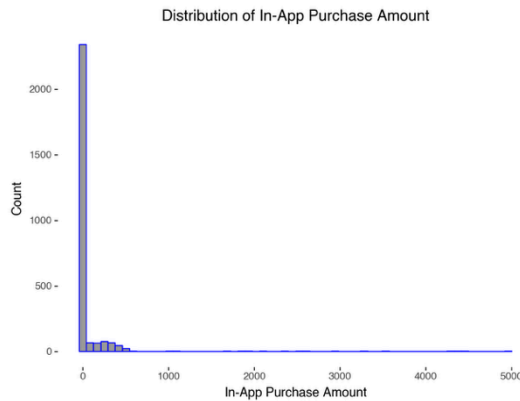


Figure 1: Distribution of In-App Purchase Amount

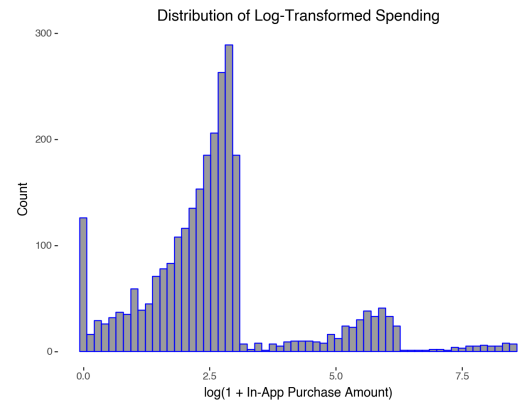


Figure 2: Log-Transformed Spending Distribution

1. Spending is highly skewed.

Both Figures 1 and 2 show how extremely skewed player spending is. In the raw distribution, almost all players spend little or nothing, creating a large spike at zero, while only a small group accounts for the majority of revenue. After applying a log transformation, the long right tail shrinks, and the distribution becomes more evenly spread out. This clearer shape also makes it easier for regression models to work well, since these models improve when the dependent variable is not heavily skewed.

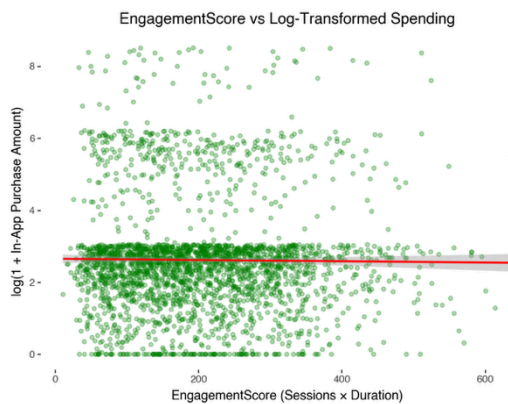


Figure 3: EngagementScore vs Log-Spending

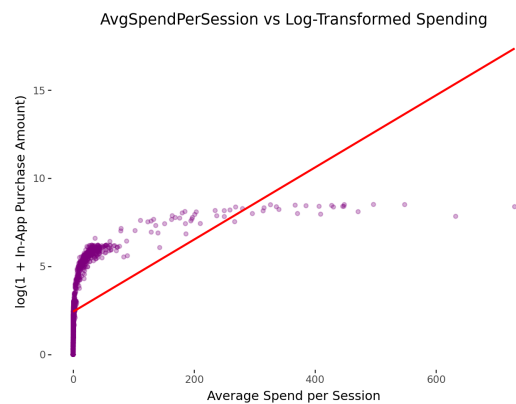


Figure 4: AvgSpendPerSession vs Log-Spending

2. Engagement is common, but spending is not.

Figure 3 shows that EngagementScore has a very weak correlation with total spending. Many players with high engagement still spend little or nothing, which is common in free-to-play games. This aligns with findings from Sifa *et al.* (2015), Hamari and Lehdonvirta (2010), and Costes and Bonnaire (2022): engagement is common, but spending is not. This means that

while higher engagement broadens the potential spending range, it does not reliably predict how much a player will spend.

3. Spending intensity matters most.

The relationship between average spend per session and total spending is strong and positively linear. Players who spend more during each gameplay session almost always become the highest overall spenders. This suggests that per-session spending behavior captures core monetization tendencies more than engagement or session frequency. However, since AvgSpendPerSession is directly based on the spending outcome itself, we consider it only as an exploratory insight and exclude it from the training data in the later modeling stage.

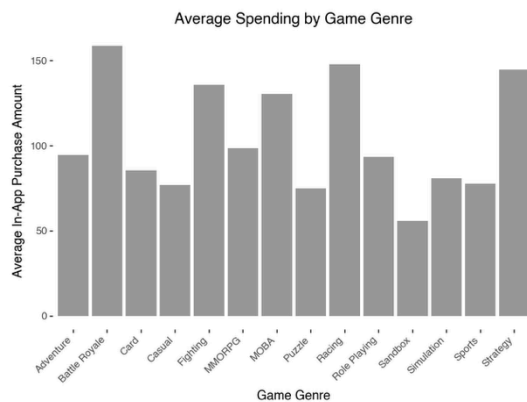


Figure 5: Average Spending by Game Genre

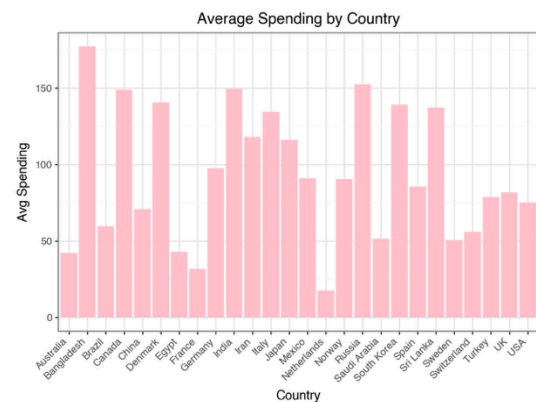


Figure 6: Average Spending by Country

4. Spending varies sharply across game genres.

This chart compares the average in-app purchase amounts across different game genres. Although the differences are not huge, some genres clearly attract higher spending than others. For example, Battle Royale, Racing, Strategy, Fighting, and MOBA games show noticeably higher average spending, which matches the fact that these genres often feature competitive mechanics, progression systems, and more opportunities for microtransactions.

5. Spending differs from country to country.

Figure 6 shows the average in-app spending across different countries. The differences are quite apparent: some countries have much higher spending levels than others. For example, Bangladesh, Canada, Denmark, India, and Russia stand out with relatively high average spending. Meanwhile, countries like Netherlands, France, Egypt, and Australia are toward the lower end of the range. These differences probably reflect variations in purchasing power, cultural attitudes toward mobile gaming, and the popularity of in-app purchases in each region.

Player Segmentation Based on Engagement and Spending



Figure 7: Player Segmentation (Casual / Regular / Whale)

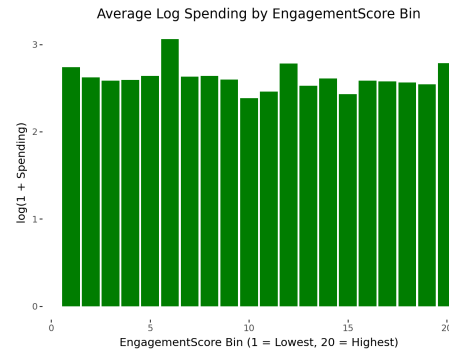


Figure 8: Average Log-Spending by EngagementScore Bins

6. Player engagement and spending vary widely across different types of users.

Figure 7 compares three player segments based on engagement and spending. Whale Users (blue) are mostly in the upper region, consistently spending a lot regardless of engagement. Regular Spenders (green) are in the middle, often with higher engagement but moderate spending. Casual Players (red) stay in the low-spending area, even when their engagement varies widely. These patterns highlight different spending behaviors and motivations across the three groups.

7. Average spending remains relatively stable across different engagement levels.

This plot compares mean log-transformed spending across 20 EngagementScore bins. Overall, the differences between bins are small, and there is no clear upward or downward trend. While a few bins (such as 6 and 12) show slightly higher averages, the variation is limited. This suggests that higher engagement does not consistently lead to higher spending, and that spending may depend more on individual motivations or purchase preferences than on engagement alone.

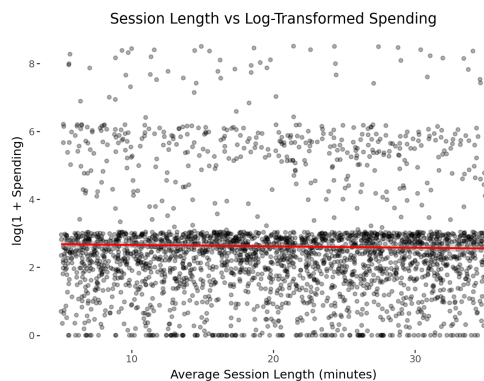


Figure 9: Session Length vs Log-Spending

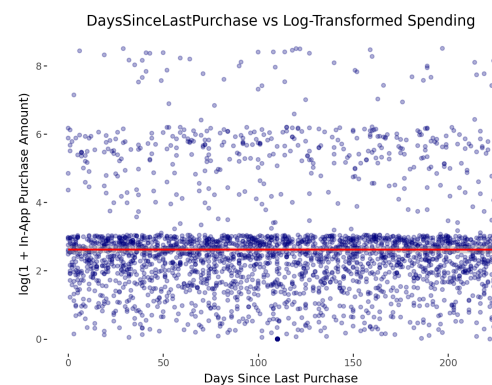


Figure 10: Recency vs Log-Spending

8. Session length shows very little relationship with player spending.

This plot compares average session duration with log-transformed spending, and the pattern remains mostly flat. Players with short sessions and those with longer sessions cluster at similar spending levels, and the regression line shows little to no upward or downward trend. Although a few high-spending players appear across different session lengths, most points fall into a narrow horizontal band, indicating that spending habits do not systematically grow with longer play sessions. These results suggest that session duration alone is not a reliable indicator of how much a player spends, and that spending behavior is likely influenced by other factors beyond how long a player remains in each session.

9. The time since a player's last purchase shows little connection to how much they spend.

This plot shows the number of days since the last purchase against log-transformed spending, and the distribution remains fairly flat across the entire range. Players who recently made a purchase and those who have not bought in over 200 days have similar spending levels, as indicated by the regression line, which shows almost no slope. While a few high-spending players appear at different points, most users cluster closely around the same spending level. These patterns suggest that how recently someone purchased is not a strong predictor of current spending, likely because spending habits are more influenced by individual preferences and specific motives than by the time since the last transaction.

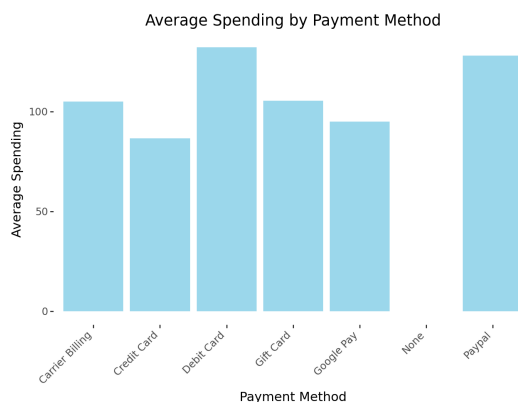


Figure 11: Average Spending by Payment Method

10. Average in-app spending differs noticeably across payment methods.

This plot compares the average spending levels for players using different payment options, and the differences are quite clear. Debit Card and PayPal users have the highest average spending, standing out from the other groups. Carrier Billing and Gift Card users also spend relatively more, while Google Pay and Credit Card users fall into the lower range. These variations may reflect differences in purchase friction, preferred spending habits, or the types of players who select each payment method.

These visualizations help clarify key patterns and guide the selection of modeling strategies.

4. Methods

To assess whether player spending can be predicted from behavioral and demographic features, I created two categories of supervised learning models: regression models to forecast the log-transformed spending amount, and a classification model to identify whale users.

For the regression task, the models included Linear Regression, Lasso Regression, Decision Trees, Random Forest, and XGBoost. This combination of linear and nonlinear methods allows for comparison between simple, interpretable baselines and more adaptable tree-based techniques that can capture thresholds, heterogeneity, and complex feature interactions in behavioral data.

For classification, Logistic Regression was chosen because of its interpretability and suitability for imbalanced outcomes, with model performance assessed through ROC/AUC.

These modeling choices are supported by practical considerations and previous research. Linear and Lasso regression offer interpretable benchmarks and help determine whether spending patterns are roughly linear relative to the engineered features. Decision Trees and ensemble methods like Random Forest and XGBoost are well-suited for skewed, high-variance behavioral data often seen in mobile-game monetization. Prior research (e.g., Chen *et al.*, 2018) demonstrates that tree-based models perform well in lifetime-value and churn prediction because they capture nonlinear effects, spending concentration, and sudden behavioral shifts. XGBoost, in particular, is effective for structured, noisy datasets and can detect subtle pattern differences that linear models cannot. SHAP values are used with XGBoost models to offer a detailed, model-agnostic explanation of which features most influence predicted spending outcomes.

Despite these advantages, several limitations still exist. First, the synthetic dataset might underrepresent real-world behavioral noise, psychological factors, and motivational drivers that influence spending. Additionally, the highly imbalanced spending distribution—many zero spenders versus a small number of whales—makes accurate individual-level prediction challenging for any model. Lastly, while tree-based models are powerful, their performance heavily relies on the presence of rich and informative features; in this dataset, behavioral signals are relatively weak, which limits overall predictive accuracy.

5. Results

This section summarizes the performance of the regression and classification models, presents the relevant error metrics, and interprets the feature importance and SHAP outputs. All models were trained on 80% of the dataset and evaluated on the remaining 20%.

5.1 Regression Model Performance

All regression models were fitted on the log-transformed spending variable and evaluated using MAE, RMSE, and R^2 on the test set.

Test-set metrics

Model	MAE	RMSE	R^2
Linear Regression	0.975061	1.450415	0.052530
Lasso Regression	0.930866	1.413263	0.100446
Decision Tree	0.937660	1.433506	0.074492
Random Forest	1.007974	1.456062	0.045137
XGBoost	1.128388	1.560326	-0.096509

Table 1: Summary of regression metrics on the test set

Across all models, R^2 values stay around 0 to 0.10, showing that only a small part of the variation in log-spending is explained by the features. Even after log-transforming the target, MAE and RMSE remain near 1, indicating that individual spending predictions are noisy.

5.1.1 Linear and Lasso regression

The baseline linear regression model achieves:

- MAE: 0.9751
- RMSE: 1.4504
- R^2 : 0.0525

Lasso regression, optimized through 5-fold cross-validation over a logarithmic grid of α values (best $\alpha \approx 0.045$), slightly enhances performance:

- MAE: 0.9309
- RMSE: 1.4133
- R^2 : 0.1004

Interpretation:

1. Player spending does not follow straightforward linear patterns; even the most accurate linear model accounts for just around 10% of the variance.
2. Lasso keeps only a small set of weak predictors, mainly country indicators and payment-method dummies, while some engineered behavioral variables are shrunk to zero.

This matches the EDA: spending behavior appears highly nonlinear and noisy, with no strong linear signal.

5.1.2 Tree-based models: Decision Tree and Random Forest

To capture potential nonlinear patterns, a Decision Tree Regressor was trained. Numeric features were preprocessed with median imputation and standardized using a ColumnTransformer pipeline, and the tree depth was limited to `max_depth = 4` to prevent overfitting. On the test set, the tree achieved:

- MAE: 0.9377
- RMSE: 1.4335
- R^2 : 0.0745

Thus, the tree explains roughly 7% of the variance which is slightly better than linear regression but still modest.

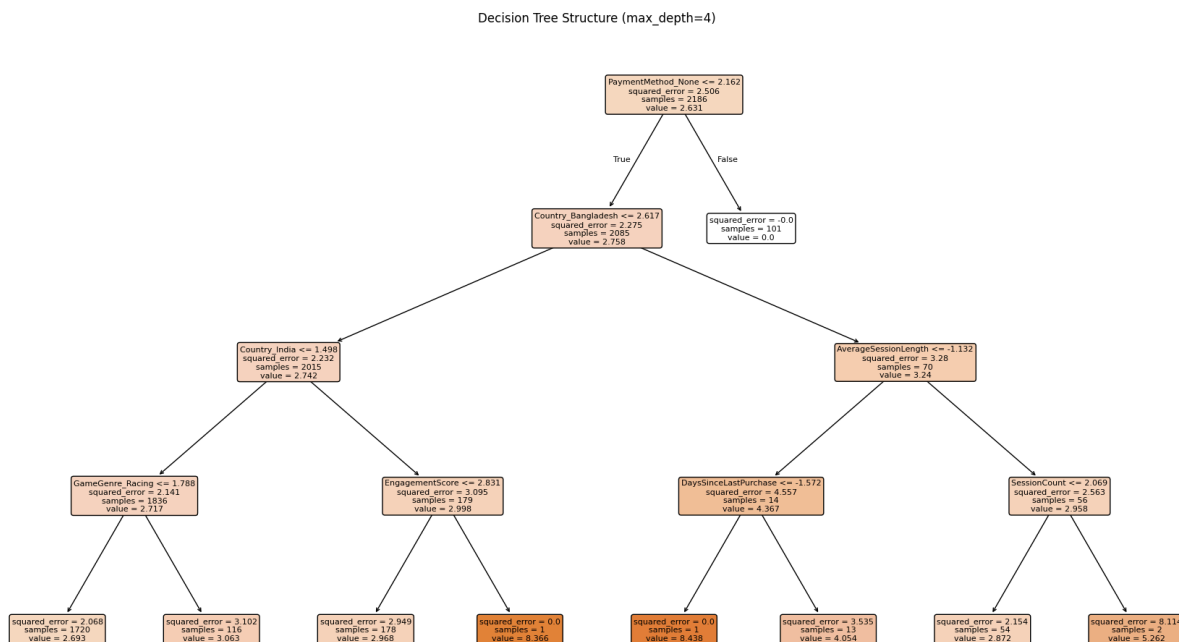


Figure 12: Decision Tree Structure

Inspection of the fitted tree shows a layered segmentation:

1. Root split on `PaymentMethod_None`: distinguishes players without a linked payment method from those with one, showing that payment setup is a key factor in whether players spend at all.
2. Country splits (e.g., Bangladesh, India) among spenders highlight significant geographic differences in spending levels.

3. Deeper behavioral splits using variables such as GameGenre_Racing, EngagementScore, DaysSinceLastPurchase, AverageSessionLength, and SessionCount, which improve predictions within payment/country segments.

Although these thresholds are intuitive, a single tree is unstable and tends to memorize local patterns, so its overall predictive power remains limited.

To improve stability, a Random Forest Regressor with 300 trees was trained. Test-set performance is:

- MAE: 1.0080
- RMSE: 1.4561
- R^2 : 0.0451

Despite its flexibility, the Random Forest still explains only about 4–5% of the variance in log-spending, similar to the linear models.

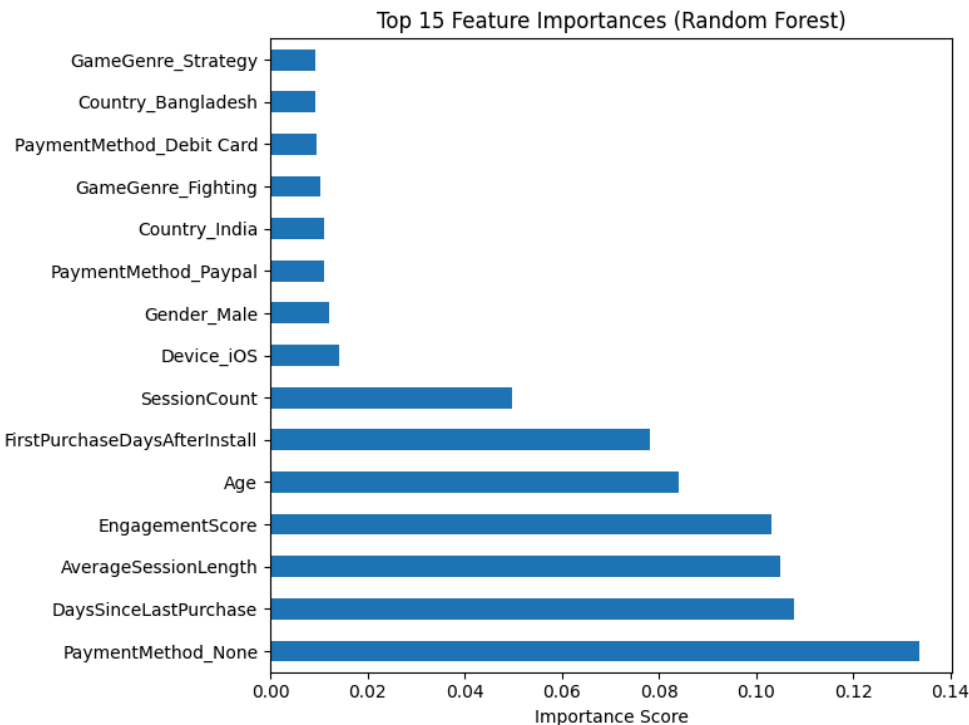


Figure 13: Top 15 Feature Importances (Random Forest)

To better understand how the Random Forest model distributes predictive influence across variables, Figure 13 ranks the top 15 predictors. The model focuses most of its importance on a few behavioral and temporal features rather than demographics or device attributes.

The feature-importance plot highlights:

- PaymentMethod_None
- DaysSinceLastPurchase
- EngagementScore
- AverageSessionLength
- Age

as the top predictors, but their importance values are small in absolute terms. This suggests that, given the current feature set, there is simply not enough signal to accurately predict individual spending levels, even when nonlinear interactions are allowed.

5.1.3 XGBoost Regression

An XGBoost regressor was trained with default parameters, resulting in the following test set performance:

- MAE: 1.1284
- RMSE: 1.5603
- R^2 : -0.0965

The negative R^2 shows that the model performs worse than just predicting the average of the log-spending. This indicates that boosting does not capture any additional structure beyond what simpler models already find.

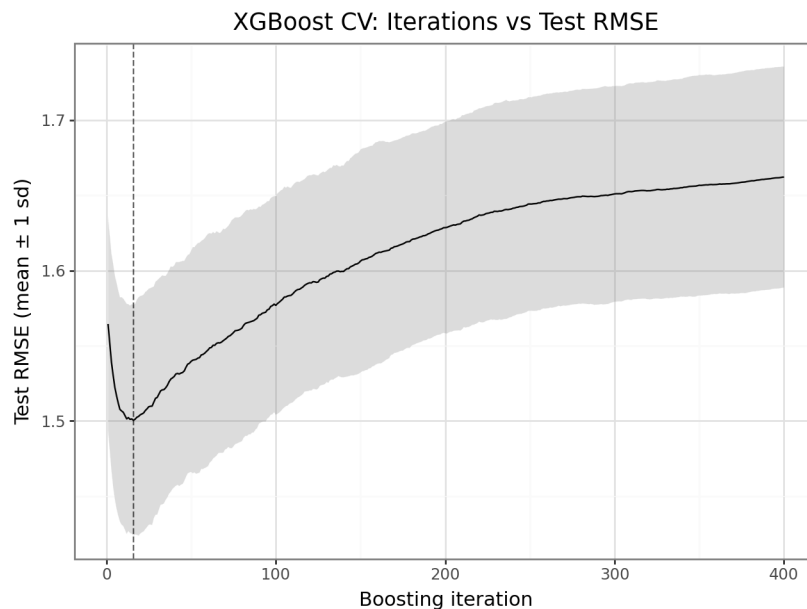


Figure 14: XGBoost CV : Iterations vs Test RMSE

To further tune the model, 5-fold cross-validation was applied. As shown in Figure 14, the test RMSE reaches its lowest point (around 1.50) at iteration 16. Beyond this point, more boosting rounds start to raise the error, indicating early signs of overfitting. Even at its optimal iteration,

performance stays similar to that of linear and tree-based models, implying that the available features lack enough additional signal for boosting to leverage.

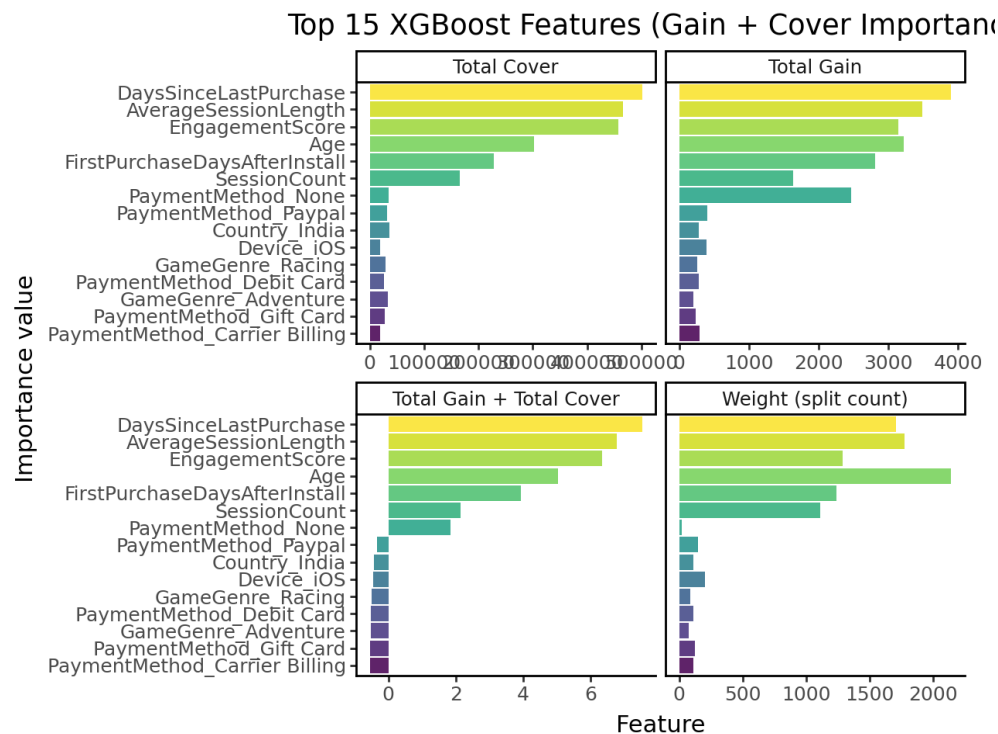


Figure 15: Top 15 XGBoost Features (Gain + Cover Importance)

To understand what XGBoost is learning, feature importance was assessed using four metrics—gain, cover, weight, and the combined total_gain + total_cover measure. Although these metrics differ in how they quantify contribution, they all emphasize a consistent group of influential variables:

- DaysSinceLastPurchase
- AverageSessionLength
- EngagementScore
- Age
- FirstPurchaseDaysAfterInstall
- SessionCount
- PaymentMethod indicators
- Country_India

These variables closely match the Random Forest rankings. Recency, engagement, and session-related behavioral features consistently rank highly, while most demographic and categorical attributes contribute very little.

Overall, these results suggest that predicting continuous spending levels is quite challenging in this dataset. The available behavioral and demographic features have limited explanatory

power, and individual spending amounts remain highly noisy and irregular, with only weak, inconsistent links to observable player attributes.

5.2 SHAP Analysis

To further interpret the XGBoost regression model, SHAP values were calculated to measure how each feature influences the predicted log-spending at both the overall and individual levels. Although the model's predictive accuracy is modest, the SHAP results show consistent and understandable behavioral patterns that match the broader findings in the regression analysis.

5.2.1 Global Importance

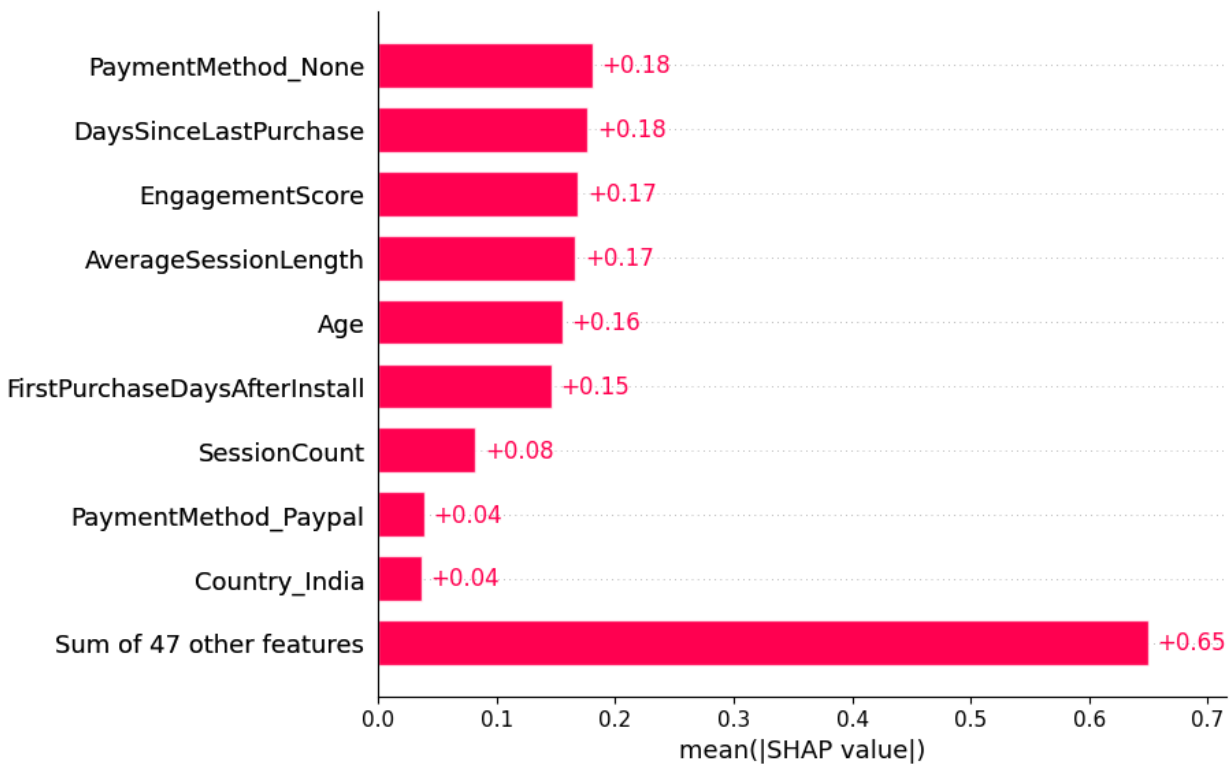


Figure 16: SHAP Summary Bar Plot

The SHAP summary bar plot displays the features with the greatest average impact on predicted log-spending are:

- PaymentMethod_None
- DaysSinceLastPurchase
- EngagementScore
- AverageSessionLength
- Age
- FirstPurchaseDaysAfterInstall
- SessionCount, with smaller effects from
- PaymentMethod_Paypal and Country_India.

These results show that the model relies primarily on payment setup, purchase recency, and engagement/session patterns when predicting. Most other categorical attributes are grouped in the “sum of 47 other features” bar, indicating they have very little overall impact.

5.2.2 Distribution of Feature Effects

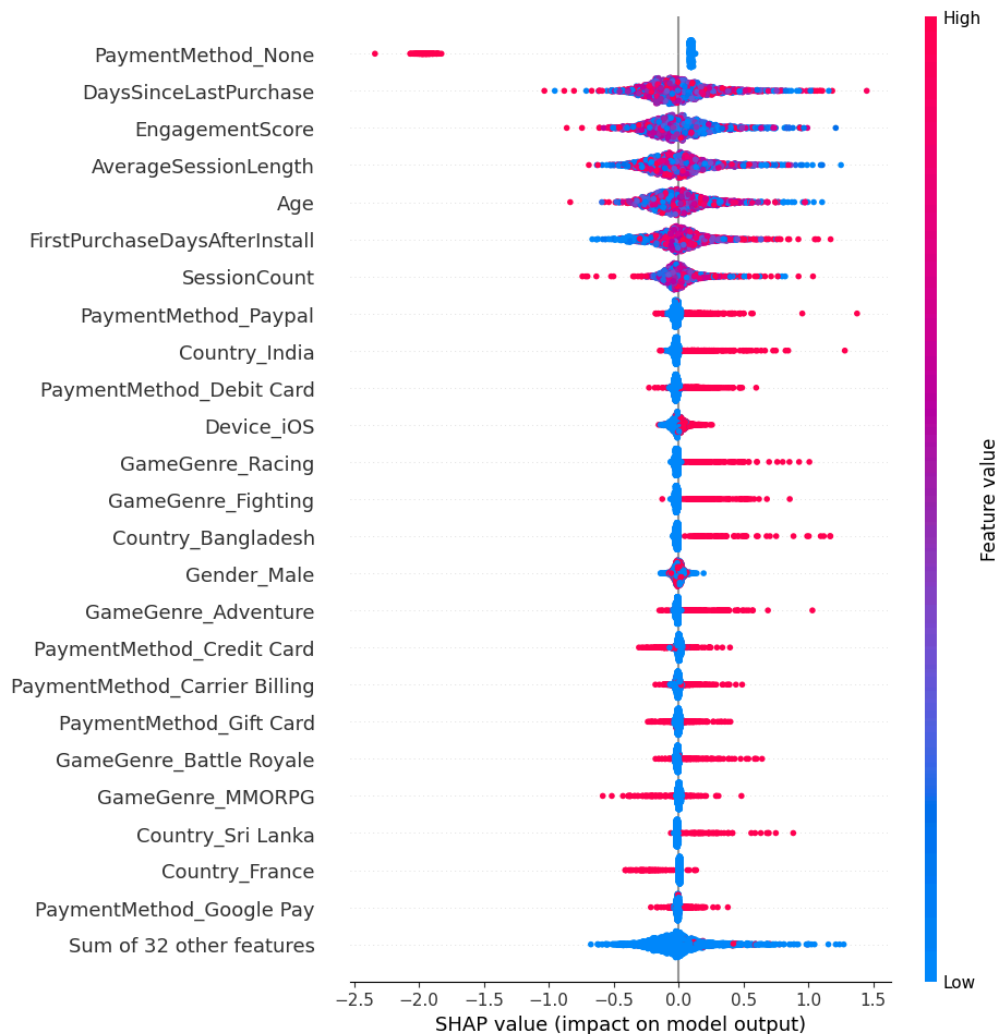


Figure 17: SHAP Beeswarm Plot

The beeswarm plot visualizes how each feature affects predictions across the entire player base:

- **PaymentMethod_None:**
No payment method on file produces negative SHAP values, lowering predicted spending. Having a valid payment method increases predictions.
- **DaysSinceLastPurchase:**
Recent buyers tend to have higher predicted spend, while long-inactive users are rated lower.

- EngagementScore, AverageSessionLength, SessionCount:
Higher values shift predictions upward, but their SHAP spread ranges are relatively narrow, indicating moderate influence.

Most other variables show SHAP values clustered near zero, indicating their negligible role in the model.

5.2.3 SHAP Waterfall Plot (Single Player)

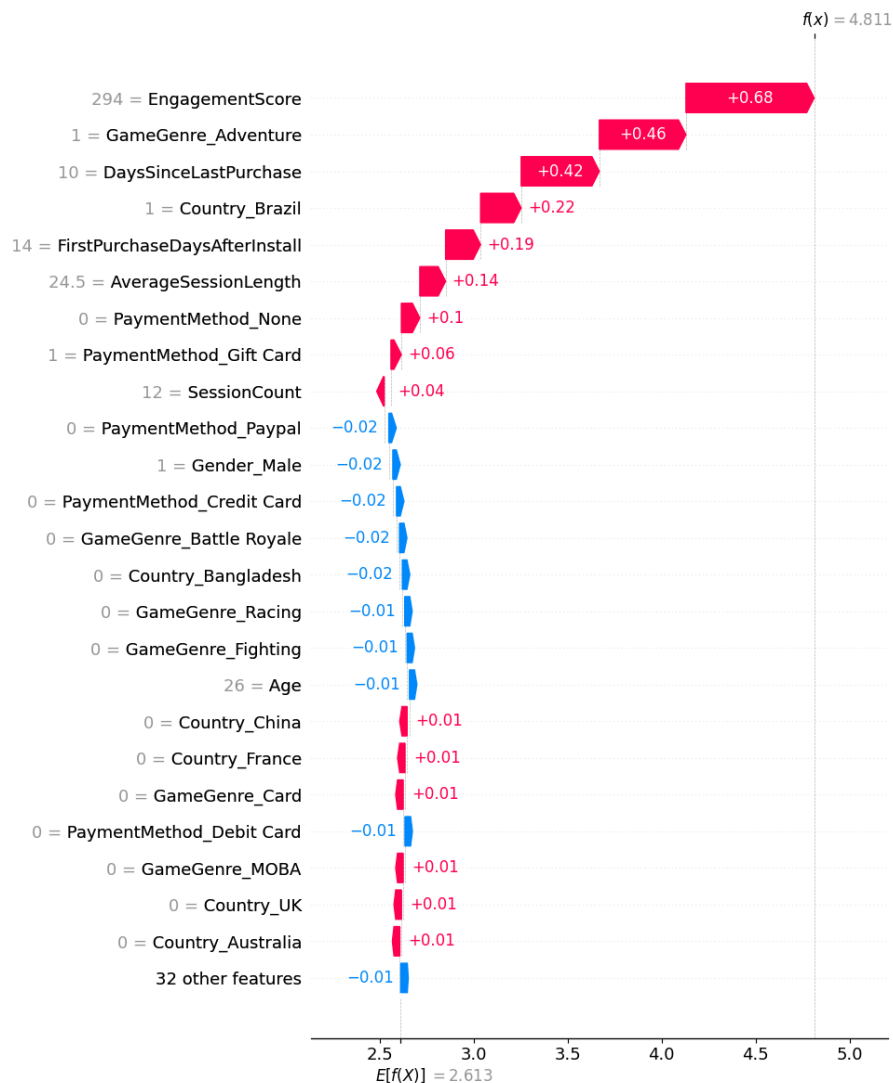


Figure 18: SHAP Waterfall Plot

The waterfall plot for a representative high-spending player shows how individual features combine to raise the predicted log-spend:

- Short DaysSinceLastPurchase,
- High EngagementScore,
- Long AverageSessionLength, and

- Having a valid payment method

All contribute positively, significantly raising the prediction above the baseline. Other variables cause only small additional adjustments. This pattern indicates that the model mainly identifies high spenders through recent activity, strong engagement, and a stored payment method.

5.2.4 SHAP Decision Plot (Multiple Players)

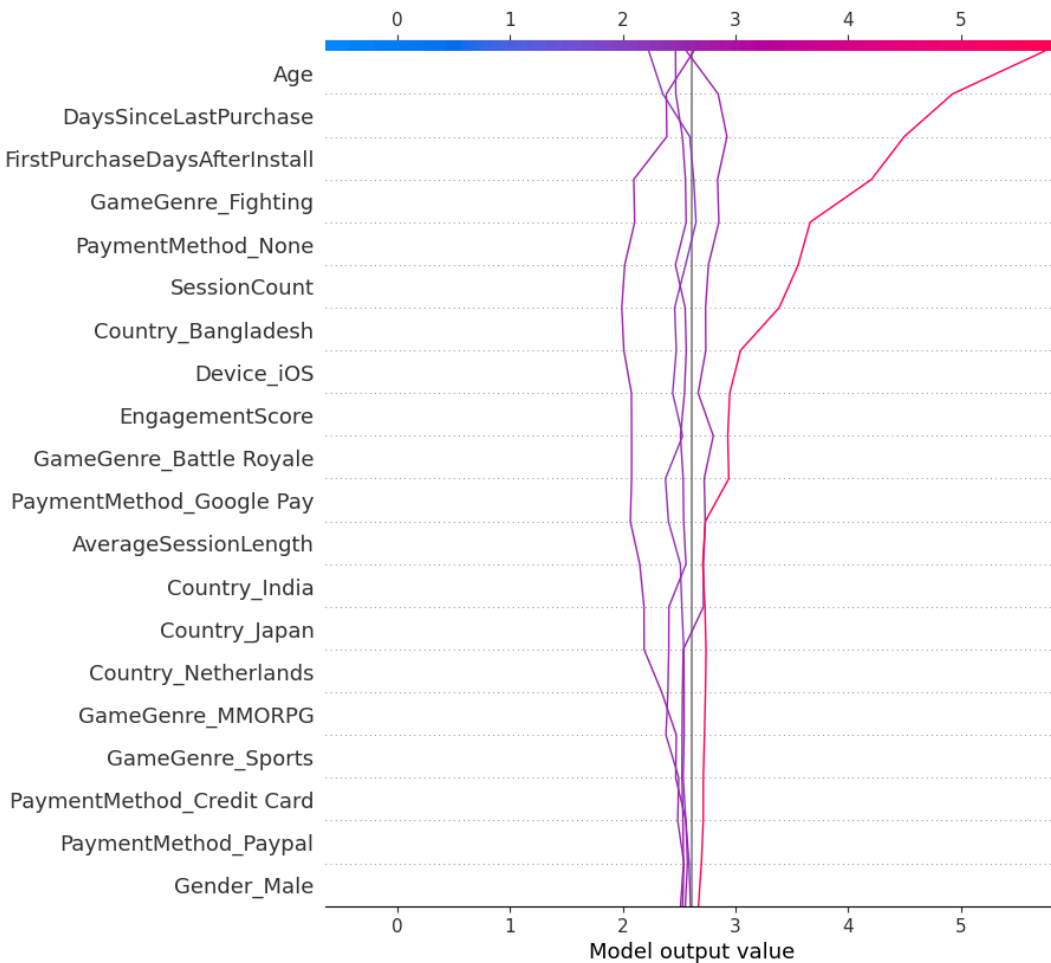


Figure 19: SHAP Decision Plot

The decision plot traces SHAP trajectories across several players:

- For higher-spending users, feature contributions accumulate upward through a combination of:
 - low DaysSinceLastPurchase,
 - earlier FirstPurchaseDaysAfterInstall,
 - stronger engagement,
 - and having a specified payment method.
- For lower-spending users, the opposite pattern emerges:
 - long inactivity,

- PaymentMethod_None,
 - lower engagement metrics
- all pull the model toward lower predictions.

This confirms that the model consistently makes decisions based on behavioral readiness and payment availability, not demographic attributes.

5.2.5 SHAP Heatmap (200 Players)

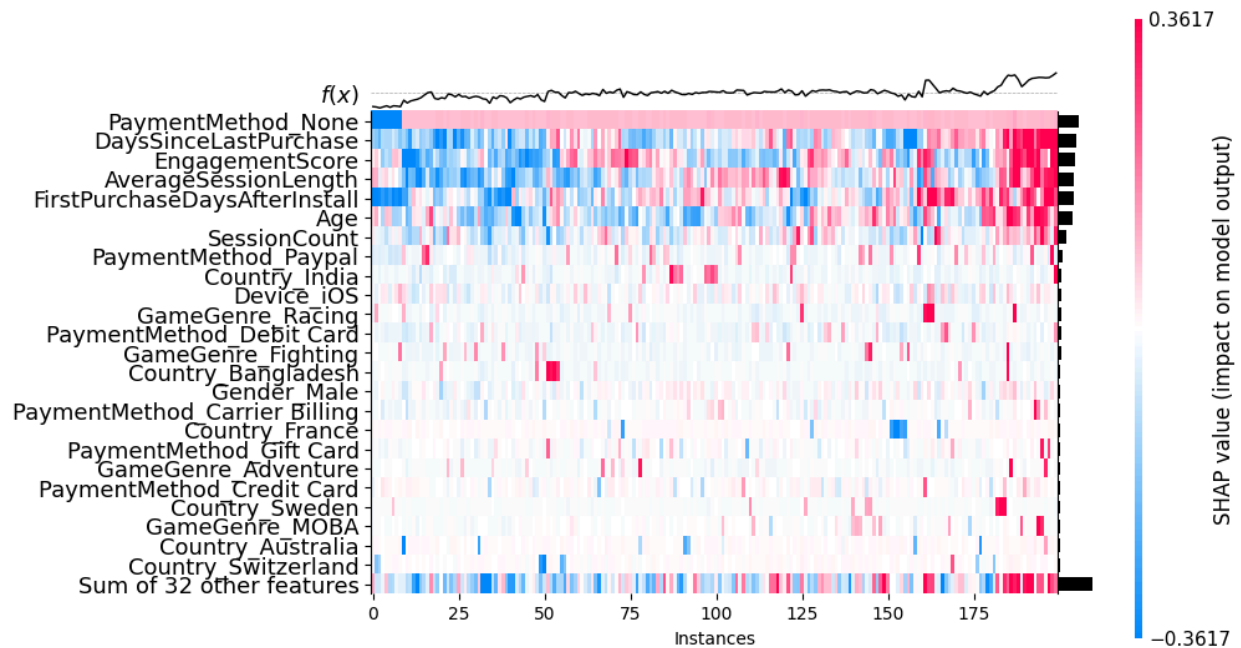


Figure 20: SHAP Heatmap

The SHAP heatmap summarizes contributions for 200 players. Distinct horizontal bands are visible for:

- PaymentMethod_None
- DaysSinceLastPurchase
- EngagementScore
- AverageSessionLength

These features display clear and consistent red/blue contrasts, suggesting a systematic influence throughout the dataset. Meanwhile, most other features show faint, centered patterns close to zero, indicating minimal effect.

The consistency of these bands across many players suggests that, despite the limited overall R^2 , the model has learned a stable and interpretable rule structure: players who are recently active, highly engaged, and have a valid payment method are predicted to spend more.

5.3 Challenge

A key challenge during model fitting was the highly noisy and imbalanced nature of spending behavior. Even after log transformation and extensive tuning, most models had difficulty generalizing—reflecting the inherent unpredictability of individual-level spending.

6. Discussion

6.1 Discussion of Model Results

The regression results collectively suggest that predicting continuous in-app spending is very challenging with the available dataset. Across all models, the test-set R^2 values stayed between 0 and 0.10, while MAE and RMSE hovered around 1, even after log-transforming the target variable. These patterns indicate that individual spending amounts are highly variable, noisy, and only weakly explained by observable behavioral or demographic attributes.

Linear and Lasso regression revealed that spending does not have strong linear relationships: even the top-performing Lasso model explained only about 10% of the variance, mostly retaining payment method and country indicators while shrinking many behavioral features to zero. This matches the exploratory findings, which show weak and inconsistent linear trends between engagement and spending.

The Decision Tree and Random Forest models identified some nonlinear patterns—mainly related to payment readiness, recency, and engagement metrics—though their predictive accuracy remained limited. Random Forest, even with 300 trees, achieved only $R^2 \approx 0.04$ – 0.05 , indicating that the underlying behavioral phenomena are complex and not easily captured by shallow nonlinear segmentation.

XGBoost, despite its flexibility, did not yield stronger performance, producing negative R^2 on the test set. Cross-validation indicated early overfitting and little benefit from more boosting iterations. This confirms that boosting cannot uncover additional predictive structure beyond what the simpler models already identify, likely because the feature set lacks strong incremental signals.

The SHAP analysis reinforces these findings. Globally, the most influential features for XGBoost align with the tree-based importance rankings. SHAP beeswarm and heatmap plots reveal that only a small subset of behavioral and temporal features consistently influence predictions, while most other variables have near-zero impact. This shows that the model has learned a stable but limited structure: predictions mainly depend on recency, engagement, and payment setup, with little additional information derived from demographics, game genres, or device attributes.

6.2 Insights Gained from the Models

Despite the limited predictive accuracy, the models offer several actionable insights into spending behavior within mobile games:

1. Spending is highly unpredictable and extremely concentrated

Spending amounts are inconsistent and vary greatly among individuals. This confirms the industry observation that whales contribute disproportionately to revenue.

2. Engagement does not strongly predict monetization

Features like SessionCount or AverageSessionLength have limited predictive value, indicating that playtime and spending are not closely linked. This is consistent with previous research showing that enjoyment and engagement do not necessarily lead to spending.

3. Payment readiness and recency are the strongest behavioral signals

Players who:

- have a payment method on file,
- made a recent purchase, or
- engaged early after installation

are consistently more likely to spend. These features repeatedly appear among the top contributors across all models and SHAP analyses.

4. Many demographic and categorical attributes have minimal effect

Age, gender, device type, and most game genres show negligible influence. Spending behavior appears to be shaped far more by real-time behavioral signals than by static user attributes.

6.3 Implications for Game Monetization Strategy

The combined regression and SHAP results point to several strategic implications for mobile-game monetization.

1. Focus on Payment Readiness as a Core Funnel Stage

PaymentMethod_None is consistently the strongest predictor across models and SHAP:

- Encourage early linking of payment methods
- Provide incentives for storing payment details
- Reduce friction in the checkout process

Players with stored payment methods are systematically more likely to spend, and spend more.

2. Use Recency-Based Interventions to Boost Spending

DaysSinceLastPurchase is among one of the most influential features:

- Trigger targeted offers shortly after a purchase
- Use inactivity-based push notifications

- Apply timed discounts to reactivate lapsed purchasers

Recent purchasers are markedly more likely to spend again.

3. Reevaluate Engagement-Based Targeting

Traditional metrics such as session length, engagement score, and session count provide only marginal predictive power:

- Engagement should guide retention strategies
- Spending should be targeted using recency, payment signals, and early behaviors
- Avoid assuming that “more playtime implies “higher spending”

This prevents wasteful targeting of high-playtime but low-spending users.

4. Tailor Promotions to Early-Purchase Patterns

FirstPurchaseDaysAfterInstall consistently contributes to predictions:

- Early spenders should be funnelled into premium bundles
- Late or non-spenders need “starter offers” and lower-friction entry points

Onboarding-phase behavior is especially informative.

5. Classification of Whale vs. Non-whale

Since spending in mobile games is highly concentrated among a small group of “whale” users, early identification of potential whales holds significant strategic importance. Precisely recognizing these users would allow studios to:

- personalize high-value offers,
- prioritize retention interventions, and
- forecast revenue more reliably.

With this motivation, a logistic regression classifier was trained to distinguish whales from non-whales.

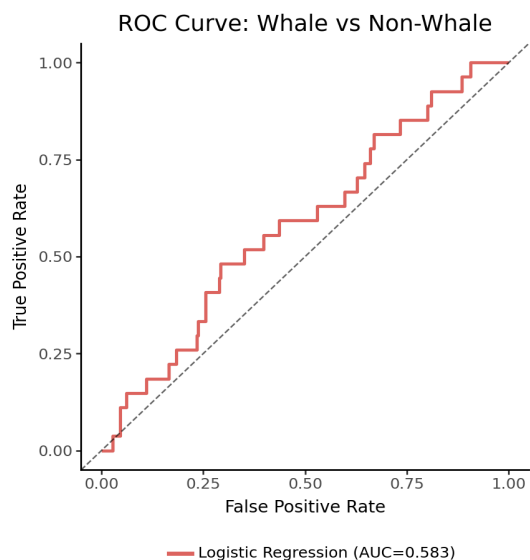


Figure 21: SHAP Heatmap

The classifier achieves an AUC of 0.583, which is just above the 0.5 diagonal line representing random guessing. The ROC curve remains close to the diagonal across all thresholds, indicating that logistic regression has limited ability to distinguish whales.

To further evaluate classification performance, a confusion matrix was computed.

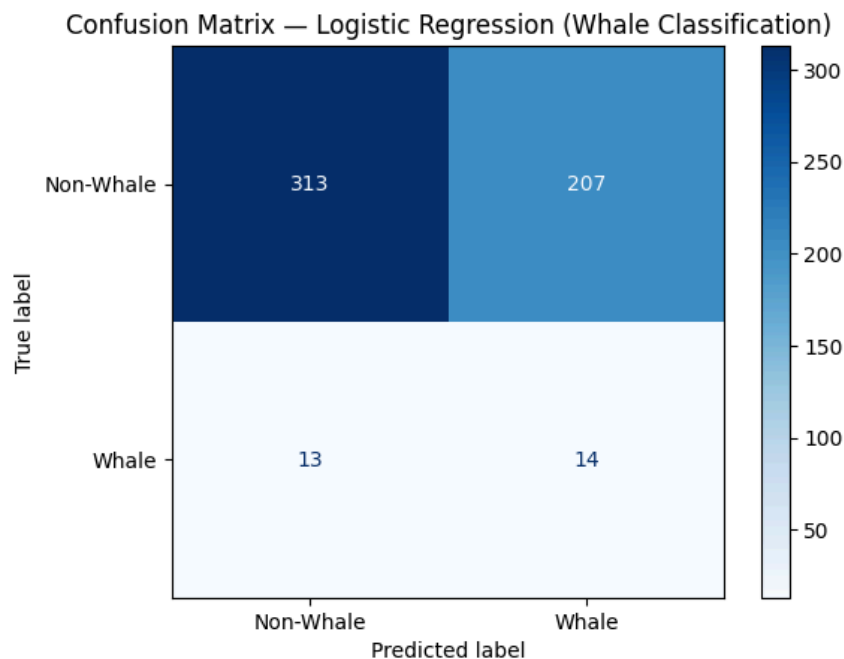


Figure 22: Confusion Matrix — Logistic Regression (Whale Classification)

As shown in Figure 22, the logistic regression model correctly identifies 313 non-whale players, but mislabels 207 non-whales as whales, indicating a substantial false-positive rate.

Among true whale users, the model identifies only 14 whales correctly, while 13 whales are classified as non-whales, resulting in a high false-negative rate.

Overall classification accuracy is approximately 0.5978, which is only slightly higher than random guessing. This modest value is expected given the severe class imbalance and the limited behavioral separability available in the dataset. In practice, because whales constitute only 5% of players, accuracy is dominated by the majority class and therefore tends to overstate actual performance—even though the model struggles to correctly identify high-value users.

Taken together, the confusion matrix, accuracy score, and ROC curve indicate that logistic regression provides only a weak signal for early whale detection. Despite the significant business value of whales, the available feature set offers limited statistical distinction between whale and non-whale behavior. Several factors contribute to this outcome:

- Feature overlap between whales and regular spenders
- Linear boundaries are too simplistic
- Missing psychological or contextual signals
- Synthetic data limitations

The ROC results further demonstrate that whale detection is far more challenging than predicting general spending levels. Although whales generate a disproportionately large share of total revenue, their early behavioral patterns are not easily distinguishable when relying solely on basic demographic and engagement metrics. This finding highlights the need for:

- Additional features such as purchase frequency, shop visits, offer views, and event participation
- Tree-based or sequential models

Identifying whales early requires richer behavioral and transactional signals than those available in this dataset.

7. Conclusion and Future work

7.1 Conclusion

This project assessed whether mobile-game player spending can be predicted using behavioral, demographic, and payment features. Across all regression models, including Linear, Lasso, Decision Tree, Random Forest, and XGBoost, the accuracy remained low, with R^2 values between 0 and 0.10. These results show that individual spending is very noisy and only weakly related to observable variables. Even the most flexible models captured little structure, indicating that spending behavior is irregular and hard to model accurately.

Despite these limitations, the analysis yielded several consistent insights. Payment readiness, especially having a stored payment method, purchase recency, and early-lifecycle behaviors proved to be the strongest signals across the tree-based models and SHAP results. In contrast, engagement metrics such as session count and average session length showed little predictive value. Demographic attributes and most game-genre indicators contributed minimally, confirming that real-time behavioral factors are far more important than static user characteristics.

The whale-classification model showed similarly weak predictive power, with logistic regression reaching an AUC of 0.583. This indicates that early whale identification is difficult with the current feature set, and that high-value spenders do not display clearly distinguishable early-behavior patterns in this dataset.

Overall, the project shows that although accurately predicting spending is challenging, the consistent behavioral patterns found, especially regarding payment setup and purchase timing, provide useful guidance for monetization strategies.

7.2 Future Work

Given more time and access to deeper data sources, several meaningful extensions could significantly improve both predictive accuracy and practical relevance:

1. Incorporate Richer Behavioral and Transactional Data

Current features (e.g., engagement scores, session length) provide only coarse behavioral signals. Future work should include:

- Purchase history sequences (frequency, size, timing of past purchases)
- In-game economic behavior (currency balance, shop visits, item browsing)
- Event participation and progression metrics
- Promotion responses (click-throughs, discounted bundle purchases)

These features would enable models to capture temporal patterns and micro-behaviors that may strongly influence monetization.

2. Apply Sequential and Deep Learning Models

Spending decisions are time-dependent. Methods such as:

- LSTMs/GRUs,
- Temporal convolutional networks, or
- Transformer-based sequence models

could capture dynamic behavioral evolution more effectively than static tabular models.

3. Improve Whale Classification with Advanced Models

The logistic regression baseline can be replaced with:

- Gradient boosting classifiers,
- Random forests,
- XGBoost with class-weighting, and
- Focal loss models

to better handle imbalanced outcomes and nonlinear separations.

4. Explore Causal or Semi-Causal Frameworks

To inform monetization strategy, future work could examine:

- Whether early purchase timing *causes* higher lifetime spending,
- How payment-method setup influences subsequent behavior,
- Which interventions effectively increase conversion.

Approaches such as uplift modeling, causal trees, or A/B test simulation would add strategic insight.

5. Validate Models Using Non-Synthetic Data

Because the current dataset is synthetic, applying the framework to real behavioral data from an operational game would help:

- Evaluate generalizability,
- Identify missing features, and
- Benchmark true predictive limits.

These directions would provide clearer predictive signals and support more detailed monetization and retention strategies in real mobile-game environments.

Dataset:

https://drive.google.com/file/d/1pV9_ag0MFyHQhqNJpbEM6SJRn_5bVHtr/view?usp=drive_link

Coding:

https://drive.google.com/file/d/1YwoojRZbaFcMtALh9vHf5YyuUXel3be7/view?usp=drive_link

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